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St. Joseph's College of Arts & Science (Autonomous)

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B. Sc DEGREE EXAMINATION APRIL 2021

ASCS402 - STATISTICAL METHODS FOR COMPUTER APPLICATIONS – II

Time : 3 hrs **Max Marks :75**

SECTION - A (10x2 =20)

Answer ALL the questions

1. Define Probability mass function of Random variable.
2. Write down a two-way classification the analysis of variance table.
3. Find the expectation of the number on a die when thrown.
4. $f(x) = \begin{cases} kx, & 0 \le x \le 1 \\ ak, & 1 \le x \le 2 \\ -kx+3k, & 2 \le x \le 3 \\ 0, & elsewere \end{cases}$ Determine the constant k
5. Define : Binomial Distribution
6. X is normally distributed and the mean of X is 12 and S.D is 4. Find out the probability of $X \geq 20$.
7. A sample of 400 male students is found to have a mean height of 171.38cm. Can it be reasonably regarded as a sample from a large population with mean height 171.17cm and standard deviation 3.30 cm?
8. A random sample of 27 pairs of observations from a normal population gives a correlation coefficient of 0.42. Is it likely that the variables in the population are uncorrelated?
9. What is Randomized Block Design?
10. Define One way classification.

SECTION - B (5 x 5= 25)

Answer ALL the Questions

11. (a) Two dice are tossed. Find the probability of getting an even number on the first die or a total of 8.

(Or)

(b) The contents of urns I, II, III are as follows.

1 white, 2 black, and 3 red balls
2 white, 1 black and 1 red balls and
4 white, 5 black and 3 red balls. One urn is chosen at random and two balls drawn from it. They happen to be white and red. What is the probability that they come from urns I, II or III?

12. (a) If the joint distribution function of X and Y is given by:

$$F_{XY}(x,y) = \begin{cases} 1-e^{-x}-e^{-y}+e^{-(x+y)} \; ; \; x>0, y>0 \\ 0 \qquad\qquad\qquad ; \; elsewhere \end{cases}$$

(Or)

(b) State and prove Addition theorem of expectation

13. (a) X is normally distributed and the mean of X is 12 and S.D is 4. Find x_0' and x_1', when $P(x_0' < X < x_1') = 0.50$ and $P(X > x_1') = 0.25$

(Or)

(b) The mean and variance of binomial distribution are 4 and 4/3 respectively. Find $P(X \geq 1)$

14. (a) In a sample of 8 observations, the sum of squared deviations of items from the mean was 84. 4. In another sample of 10 observations, the value was found to be 102. 6. Test whether the difference is significant at 5% level. You are given that at 5% level, critical value of F for $v_1 = 7$ and $v_2 = 9$ degrees of freedom is 3.29 and for $v_1 = 8$ and $v_2 = 10$ degrees of freedom , its value is 3.07

(Or)

(b) Is a correlation coefficient of 0.5 significant if obtained from a random sample of 11 pairs of values from a normal population? Use t – test.

15. (a) The three samples below have been obtained from normal population with equal variances. Test the hypotheses that the sample means are equal.

8	7	12
10	5	9
7	10	13
14	9	12
11	9	14

The table value of F at 5% level of significance for $v_1 = 2$ and $v_2 = 12$ is 3.88

(Or)

(b) In a laboratory experiment, two random samples gave the following results:

Sample	Size	Sample mean	Sum of squares of deviations from the mean
1	10	15	90
2	12	14	108

Test the equality of sample variances at 5% level of significance. Table value of F for $v_1 = 9$ and $v_2 = 11$ is 2.90 and for $v_1 = 11$ and $v_2 = 9$ is 3. 10.

SECTION - C (3 x10 = 30)

Answer any THREE Questions

16. The joint p.d.f of two random variables X and Y is given by

$$f(x,y) = \frac{9(1+x+y)}{2((1+x)^4(1+y)^4} \; ; 0 \le x < \infty, \; 0 \le y < \infty$$

Find the marginal distribution of X and Y, and the conditional distribution of Y for X = x.

17. The joint probability distribution of two random variables X and Y is given by P(X = 0, Y = 1) = 1/3 , P (X =1, Y = -1)1/3, and P(x = 1, Y = 1) = 1/3 . Find (i) Marginal distributions of X and Y and (ii) the conditional probability distribution of X given Y = 1.

18. If X and Y are independent Poisson variates, show that the conditional distribution of X given X + Y is binomial.

19. A movie producer is bringing out a new movie. In order to map out his advertising campaign, he wants to determine whether the movie will appeal most to particular age group or whether it will appeal equally to all age groups. The produces takes a random sample from persons attending preview of the new movie, and obtains the following results.

	Age Groups				
	Under 20	**20 -39**	**40 – 59**	**60 & over**	**Total**
Liked the movie	146	78	48	28	300
Disliked the movie	54	22	42	22	140
Indifferent	20	10	10	20	60
Total	220	110	100	70	500

20. To assess the significance of possible variation in performance in a certain test between the convent schools of a city, a common test was given to a number of students taken at random form the senior fifth class of each of the four schools concerned .The results are given below. Make an analysis of variance of data.

A	**B**	**C**	**D**
8	12	18	13
10	11	12	9
12	9	16	12
8	14	6	16
7	4	8	15

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